

# Sandesh Nepal

✉ sandeshnepal106@gmail.com ☎ 9564570432 📧 sandeshnepal106 🌐 sandeshnepal106 🔗 sandeshnepal.vercel.app

## EDUCATION

**National Institute of Technology, Sikkim**  
*B.Tech in Electrical and Electronics Engineering*  
• Current CGPA: 7.30

Aug 2023 – Present

**Jawahar Navodaya Vidyalaya**  
• Intermediate: 89.83%  
• Matriculation: 92.80%

Aug 2019 – Mar 2023

## PROFESSIONAL EXPERIENCE

**Solar Quest LLP** 🌐

*Solar PV Engineer Intern*

Jun 2026 – Jul 2026  
Chennai, India

- Worked on Solar PV and electrical design, covering inverters, transformers, cables, equipment ratings, protection, and SLDs.
- Performed PVsyst simulations with 3D shading, loss analysis, and energy-yield estimation.
- Used ETAP for system modelling, including buses, transformers, breakers, and fault/protection analysis.
- Conducted site visits to Wimco Nagar Depot and Anna Nagar Tower Metro, observing PV layouts, equipment, testing, maintenance, cabling, and operations.
- Developed Python scripts to automate solar-generation data extraction and monitoring via Growatt, Trackso, iSolar, and Inverdis.

**Ririana Innovations** 🌐

*Systems Developer*

Dec 2025 – Feb 2026  
Remote

- Developed a mobile frontend with distinct, specialized interfaces for both users and administrators.
- Built a fully responsive UI by translating design prototypes into a functional mobile experience.
- Managed the application bundling process to generate APKs for testing.
- Implemented core functionality for property browsing, listing management, and booking workflows.

**Global Trendz** 🌐

*Web Development Intern*

Jun 2025 – Jul 2025  
New Delhi, India

- Contributed to the development of frontend features using React.js, HTML, and CSS.
- Assisted in debugging and fixing minor UI/UX issues, helping to maintain code quality in the production environment.
- Implemented responsive design updates to ensure web pages rendered correctly across different devices.
- Participated in code reviews, gaining exposure to professional software development workflows.

## PROJECTS

**Mighty Mouse: ESP32-Powered 4WD Combat Robot** 🌐

Mar 2026 – Mar 2026

- Designed a 4WD metal-chassis combat robot featuring a high-RPM spinner weapon.
- Programmed dual ESP32s using ESP-NOW for low-latency, peer-to-peer wireless control.
- Engineered LiPo power distribution with buck regulation, relay isolation, and inline fuses.
- Integrated high-power motor drivers and a dedicated ESC for tank steering and weapon control.

**IIoT-Based Predictive Maintenance System**

Feb 2026 – Present

- Simulated an IIoT predictive maintenance system to detect equipment faults and minimize downtime.
- Modeled a virtual ESP32 edge device and an MQTT pipeline for real-time telemetry streaming.
- Visualized and analyzed machine health metrics, including temperature and voltage fluctuations under simulated degradation.
- Tools used: Wokwi, C++, MQTT, Python, Google Colab

**IoT-Based Environmental Monitoring System** 🌐

Jun 2025 – Jul 2025

- Simulated a multi-sensor system (DHT22, MQ-135, LDR) using ESP32 for real-time environmental data.
- Implemented sensor calibration, ADC signal processing, and moving-average filtering for accuracy.
- Integrated with ThingSpeak cloud to enable live dashboards, alerts, and data visualization.
- Demonstrated applications in smart homes, pollution monitoring, and weather tracking.

**PMSG Based Gearless Variable Speed Wind Turbine System Simulation**

Feb 2025 – Mar 2025

- Simulated a gearless, variable-speed wind turbine system to enhance power generation efficiency.
- Modeled a Permanent Magnet Synchronous Generator (PMSG) and a back-to-back converter system for stable power output.
- Visualized and analyzed system performance, including grid voltage, current, and power waveforms.
- Tools used: MATLAB, Simulink

## SKILLS

**Hardware Design and Simulation**

MATLAB/Simulink, ETAP, PVSyst, Scilab, Tinkercad, Arduino IDE, ESP32, ThingSpeak, Wokwi

**Programming Languages**

C, C++, Python

**Databases**

MongoDB, MySQL

**Software Development**

HTML, Tailwind CSS, JavaScript, React.js, Node.js, Express.js, PHP, React Native

**Data Analysis & Visualization**

MS Excel, Power BI

**Version Control and Hosting**

Git, Github, BitBucket, Render, Vercel